Megha Neelamana

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PROFESSIONAL SUMMARY

AI and software engineer with an MSc in Data Science and Artificial Intelligence and experience building full-stack AI products, agentic workflows, and scalable backend systems. Skilled in Python, TypeScript, cloud infrastructure, machine learning, and observability, with a track record of shipping production features as a sole developer and early team member.

EXPERIENCE

HyperionWave Sep 2025 - Present
AI Engineer
- Design and implement advanced logic and algorithms using Model Context Protocol (MCP) to enhance the decision-making accuracy and autonomous behavior of AI agents.
- Architected and developed a custom AI Observability Dashboard that analyzes and identifies patterns in MCP tool calls, enabling proactive detection of logic loops and anomalous agent behaviors.
- Collaborate on the end-to-end testing and optimization of AI product features, utilizing pattern-based insights to design solutions that prevent system failures.
- Refine AI reasoning frameworks to ensure high-performance execution of complex tool-calling tasks within the product ecosystem.

Hedgwork Jan 2025 - Aug 2025
Software Engineer, AI Engineer
- Spearheaded the design and integration of full-stack AI solutions for a specialized HR platform, including AI-powered assistants and NLP-driven chatbots.
- Developed and maintained RESTful APIs and backend microservices using Python, Node.js, and Azure Cloud.
- Managed and optimized large-scale database architectures using Azure SQL and DynamoDB, ensuring high availability and query performance.
- Bridged the gap between frontend and backend teams to deliver seamless, scalable software experiences for enterprise clients.

Shink AI Tech Dec 2023 - Jul 2024
Software Engineer
- Engineered critical backend and frontend systems for a SaaS platform and mobile dating application.
- Integrated AWS Rekognition to develop a robust user verification and biometric identity tool.
- Implemented automated CI/CD pipelines and containerized microservices via Docker, significantly improving deployment frequency and system stability.
- Orchestrated data synchronization across MongoDB and DynamoDB instances, maintaining 100% data integrity across distributed services.

PROJECTS

DeployPilot | Kubernetes Application Deployment Platform
https://github.com/meghakn03/DeployPilot.git
Simplifies Kubernetes application deployment with a Go dashboard and HTTP API. Custom Kubernetes controller reconciles Deployments and Services; bearer-token authentication and namespace/image allowlists; Prometheus metrics, Grafana dashboard, and Kustomize manifests.

Folio | Financial Document Intelligence
https://github.com/meghakn03/Folio.git
Helps users find evidence in company filings and financial reports. PDF/HTML ingestion with source-linked passages; RAG using FAISS with hybrid semantic and keyword retrieval; retrieval quality measured against labeled passages.

LedgerLens | Personal Finance Transaction Monitor
https://github.com/meghakn03/LedgerLens.git
Helps reviewers investigate unusual synthetic card transactions through a local fintech dashboard. Rules flag large purchases and rapid activity with review notes; Kubernetes/Helm deployment; health checks, observability, CI, and rollback guidance.

LoopLab | ML Experimentation Platform
https://github.com/meghakn03/LoopLap.git
Enables controlled experiments on a Fashion-MNIST image classifier. PyTorch training with a rule-based agent testing one configuration change at a time; validation results compared with the best run; React dashboard, FastAPI, and SQLite experiment tracking.

TraceScope | Agent Trace Replay and Evaluation Workbench
https://github.com/meghakn03/TraceScope.git
Helps developers inspect and evaluate recorded AI-agent behavior without live model calls. React/TypeScript interface with a Python/FastAPI backend; searchable traces with loop, retry, latency, and error alerts; deterministic replay, baseline evaluation, and SQLite/JSON trace storage.

FilingFlow | SEC Filings Pipeline & Data Lakehouse
https://github.com/meghakn03/FillingFlow.git
Orchestrates SEC filing ingestion, validation, deduplication, and XBRL metric extraction with Apache Airflow; PostgreSQL serving tables and a Parquet/DuckDB lake with bronze, silver, and gold layers; retries, backfills, task logs, failure alerts, and a live dashboard; Docker Compose and GitHub Actions CI.

AeroControl | Ground Control Station Simulator
https://github.com/meghakn03/AeroControl.git
Simulates vehicle telemetry and mission commands with a Rust/Tokio backend and Flutter/Dart dashboard; gRPC and Protocol Buffers communication, health alerts, fault detection, automatic reconnection, and event logging; unit and integration tests, Docker Compose, and GitHub Actions CI/CD.

GPU Fleet Lab | GPU Cluster Management Platform
https://github.com/meghakn03/GPUFleetLab.git
Manages 16 simulated GPUs across four nodes with React and FastAPI; Kueue quotas, borrowing, priority preemption, and node-failure recovery; kind/Podman and KWOK deployment, Prometheus/Grafana monitoring and alerts, Helm automation, and integration tests.

QubitGuard | Quantum Error Correction Simulator & Decoder Benchmarking Platform
https://github.com/meghakn03/QubitGuard.git
Simulates noisy repetition and surface-code memories with Stim; compares PyMatching MWPM and baseline decoders using seeded distance sweeps and confidence intervals. React/TypeScript dashboard, FastAPI, SQLite, Plotly; pytest, Docker Compose, and GitHub Actions CI.

FrameSight | Video Object Detection & Tracking Dashboard
https://github.com/meghakn03/FrameSight.git
Detects and tracks objects in uploaded videos and live webcam streams using Python, OpenCV, YOLO (Ultralytics), and ByteTrack; confidence-labeled bounding boxes and unique track IDs; Streamlit dashboard with category counts, activity charts, and Pandas CSV exports.

BioForge Lab | Computational Biology & AI Experimentation Platform
https://github.com/meghakn03/BioForgeLab.git
Analyses DNA/protein FASTA sequences with Biopython and predicts protein solubility on public eSOL data using scikit-learn; baseline comparisons and bootstrap intervals; React/TypeScript dashboard, FastAPI, PostgreSQL, MLflow experiment tracking, Docker Compose, and pytest.

EDUCATION

University of Liverpool | 2024 - 2025
MSc in Data Science & Artificial Intelligence, Data Science

Yenepoya University | 2020 - 2023
BSc in Forensic Science, Data Analytics & Cybersecurity, Forensic Science GPA: 8.83 / 10

SKILLS

Model Context Protocol (MCP), Agentic Workflows, LLM Logic Design, NLP, Pattern Recognition, Behavioral Analytics, AWS Rekognition, Python, JavaScript, TypeScript, C++, Java, Node.js, Express.js, Flask, RESTful APIs, GraphQL, WebSockets, Microservices, React.js, React Native, Next.js, Angular, Dashboard Development, Azure SQL, DynamoDB, MongoDB, MySQL, SQL, Data Lake, AWS (Lambda, S3, ECS), Azure (AI Foundry, SQL), Docker, CI/CD (GitHub Actions, TeamCity), Serverless Architecture, Agile/Scrum, Git, Technical Documentation, System Design, Go, Kubernetes, Helm, Kubernetes Controllers, Kustomize, kind, Prometheus, Grafana, PyTorch, Machine Learning, FastAPI, SQLite, RAG, FAISS, Semantic Search, Hybrid Retrieval, Retrieval Evaluation, Financial Document Processing, Transaction Monitoring, Experiment Tracking, Trace Replay, AI Observability, JSON, Credential Redaction, Apache Airflow, ETL/ELT Pipelines, Workflow Orchestration, PostgreSQL, DuckDB, Apache Parquet, Lakehouse Architecture, Medallion Architecture, Data Modeling, Data Validation, Deduplication, Idempotent Processing, Historical Backfills, Pipeline Monitoring, Webhooks, SEC EDGAR, XBRL, Docker Compose, Bash, Unit Testing, Integration Testing, Rust, Tokio, Flutter, Dart, gRPC, Protocol Buffers, Podman, KWOK, Kueue, GNU Make, pytest, Stim, PyMatching, NumPy, Tailwind CSS, Plotly, Quantum Error Correction, Decoder Benchmarking, Computer Vision, OpenCV, YOLO (Ultralytics), ByteTrack, Streamlit, Pandas, Object Detection, Object Tracking, Video Processing, Real-Time Inference, Biopython, scikit-learn, MLflow, SQLAlchemy, FASTA Sequence Analysis, Protein Feature Extraction, Protein Solubility Prediction